

# ZHIYU WANG

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Agents    LLMs    AI4Bio/Health    Geometric DL



## 🏛️ EDUCATION

### University of Cambridge

Advanced Computer Science, Master of Philosophy (Distinction, Rank: 8/60)

Cambridge, UK

Oct 2024 — Jul 2025

– Thesis awarded the Highly Commended MPhil Project Prize 2024–2025 by the Department of Computer Science and Technology

### University College London

Computer Science, Bachelor of Science (First Class Honors)

London, UK

Sep 2021 — Jun 2024

## 🔬 RESEARCH EXPERIENCE

### Auto-Research for the Virtual Cell Challenge | SAPIENT INTELLIGENCE & SJTU

Remote

Project lead | *Collaborator: Prof. Liang Hong*

Aug 2026 — Present

– Leading a joint project with Shanghai Jiao Tong University (SJTU) that applies Praxist, our AI-scientist framework, to zero-shot prediction of CRISPRi knockdown responses in the [Virtual Cell Challenge 2026](#); ranked 198/1,304 (top 15.2%) on the preliminary leaderboard.

### Bayesian Optimization for Auto-Research Systems | MILA – QUEBEC AI INSTITUTE

Remote

Collaborator | *Supervisor: Prof. Jian Tang*

Apr 2026 — Present

– Integrated Bayesian optimization into agentic evolution: a Gaussian process predicts gains; Thompson sampling picks the next experiment.  
– Achieved the best average rank on 15 AlphaEvolve tasks and the top gold-medal rate (45%) on MLE-bench Lite.

### Protein Substructure Alignment via Optimal Transport | UNIVERSITY OF CAMBRIDGE & SJTU

Cambridge, UK

Project lead | *Supervisor: Prof. Pietro Liò & Prof. Liang Hong*

Apr 2025 — Sep 2025

– Developed PLASMA, a plug-and-play optimal-transport module for local substructure alignment on any protein language model. ([ICLR26](#))  
– In detecting shared substructures, PLASMA reached ROC-AUC 0.96–0.99 vs. 0.81–0.93 for the widely used global aligner TM-align at  $\sim 50\times$  its speed, and outperformed the state-of-the-art local method EBA at  $\sim 3\times$  its speed, with much cleaner alignments.  
– Showed with PyMOL case studies that the aligned substructures are structurally similar, not artifacts of sequence similarity.

### Hierarchical Protein Representation Learning | UNIVERSITY OF CAMBRIDGE

Cambridge, UK

Master's thesis | *Supervisor: Prof. Pietro Liò*

Oct 2024 — Jul 2025

– Built a complete SE(3)-equivariant framework that adds a secondary-structure level to protein representations via a hierarchical hypergraph, allowing a coarser-grained representation that gives better global understanding. ([Invited Talk, Cambridge](#)) ([arXiv](#))  
– Leveraged this new structural hierarchy to achieve the best SCOP fold classification accuracy (43.3% vs. 38.4% for the same network without it; 40.1% best baseline), confirming the improved global understanding of protein structure.

### Multi-Omics Data Integration with GNNs | UNIVERSITY OF CAMBRIDGE

Cambridge, UK

Project lead | *Supervisor: Prof. Pietro Liò*

Feb 2025 — Sep 2025

– Built MoRE-GNN, a heterogeneous graph autoencoder learning relational graphs from single-cell RNA, protein and ATAC data; outperforms MOJITO on strongly correlated CITE-seq data (BM-CITE ARI 0.892 vs 0.868) and enables cross-modal prediction. ([arXiv](#))

### Graph-based Pain Estimation from Facial Expressions | UNIVERSITY OF CAMBRIDGE

Cambridge, UK

Project lead | *Supervisor: Prof. Hatice Gunes*

Oct 2024 — Jan 2025

– Proposed GraphAU-Pain, combining a ResNet-50 backbone with a facial action unit (AU) graph for interpretable pain estimation; transfer learning for domain shift and missing labels raised average AU F1 on UNBC from 20% to 63%. ([IJCAI25 Workshop](#)) ([IEEE CEM](#))

### Readmission Prediction with Remote Patient Monitoring Data | UNIVERSITY COLLEGE LONDON

London, UK

Undergraduate thesis | *Supervisor: Prof. Ivana Drobnjak*

May 2023 — Jun 2024

– Built readmission-prediction models (ROC-AUC 0.79 $\pm$ 0.03) from remote patient monitoring data with UCL Hospital clinicians, using SHAP to show which clinical features drive each prediction and patient-independent bootstrapping to cope with only 55 patients.

## INDUSTRIAL EXPERIENCE

### Autonomous Research Agents | SAPIENT INTELLIGENCE

ML Research Scientist

Beijing, China

Sep 2025 — Present

- Co-first author of **Praxist**, an autonomous AI-scientist framework with DAG-based execution lineage; on the 75-task MLE-bench, it won 60 medals (49 gold) vs. 55 (34 gold) for Claude Code with Opus, at about 1/12 of the cost.
- Showcased Praxist at **AI4 2026** through live booth demos and conversations with prospective customers and collaborators.

### Healthcare LLM Knowledge Distillation | IFLYTEK HEALTHCARE

Intern ML Researcher

Hefei, China

Jul 2024 — Sep 2024

- Distilled a teacher LLM into synthetic medical instruction data to fine-tune the model behind iFLYTEK's **Xiaoyi** AI health-assistant app, raising instruction-following accuracy from 34% to 82%.

### RAG-Based Financial Intelligence System | LUOJIN DATA INFORMATION

Intern ML Engineer

Shanghai, China

May 2023 — Aug 2023

- Built a RAG pipeline (MongoDB, LangChain) for financial report classification with 97% entity matching.

### Homepage Navigation Assistant for Hospitals | UCL & NATIONAL HEALTH SERVICE

Industrial Collaboration Program

London, UK

Sep 2022 — May 2023

- Built a system that crawls a hospital's website and auto-generates an assistant that answers visitor questions and guides them to the right pages, cutting setup from weeks of manual development to 30 minutes; demoed at Great Ormond Street Hospital ([media coverage](#)).

## PUBLICATIONS

Conference, Journal & Workshop Papers

\* equal contribution

1. “Fast and Interpretable Protein Substructure Alignment via Optimal Transport.”

Zhiyu Wang\*, Bingxin Zhou\*, Jing Wang, Yang Tan, Weishu Zhao, Pietro Liò, Liang Hong. [ICLR 2026](#).

2. “Data-Efficient AU-Graph Transfer Learning for Consumer-Camera Pain-Aware Sensing.”

Zhiyu Wang, Han Wei Wang, Hangqian Li, Jun Jiang, Hatice Gunes, Yang Liu. [IEEE Consumer Electronics Magazine 2026](#).

3. “GraphAU-Pain: Graph-based Action Unit Representation for Pain Intensity Estimation.”

Zhiyu Wang, Yang Liu, Hatice Gunes. [IJCAI 2025 MiGA Workshop](#).

Preprints

4. “Praxist: From Experimental Artifacts to Solution Lineages.”

Jin Li\*, Ahmed Murtadha\*, Zhiyu Wang\*, Qiwen Chen\*, William Chen\*, Yifei Wu\*, Guan Wang\*, et al. [arXiv 2026](#).

5. “Topotein: Topological Deep Learning for Protein Representation Learning.”

Zhiyu Wang, Arian Jamasb, Mustafa Hajj, Alex Morehead, Luke Braithwaite, Pietro Liò. [arXiv 2025](#).

6. “MoRE-GNN: Multi-omics Data Integration with a Heterogeneous Graph Autoencoder.”

Zhiyu Wang, Sonia Koszut, Pietro Liò, Francesco Ceccarelli. [arXiv 2025](#).

## ADDITIONAL EXPERIENCE

- **Reviewer Experience:** NeurIPS 2026; AI4Science Workshop at NeurIPS 2025 and ICML 2026
- **Teaching Experience:** Programming Tutor at UCL (2022–23)